Assignment-18

Memory Design

Exercise 1: Design Single-Port RAM (8×8) and perform following task

Tasks:

- Use reg [7:0] mem [7:0]
- Implement **read and write operation**
- Use control signals: clk, we, addr, din, dout
- Perform write on **posedge clk**

Exercise 2: Design RAM with Read/Write Enable

Design a **16×8 RAM** with separate control signals.

Tasks:

- Inputs: we (write enable), re (read enable)
- Ensure:
  - Write only when we=1
  - Read only when re=1
- Handle **invalid condition (we=1 & re=1)**

Exercise 3: Synchronous vs Asynchronous Read RAM

Design RAM with:

- **Synchronous write**
- **Asynchronous read**

Tasks:

- Compare behavior with fully synchronous RAM
- Observe timing difference in simulation

Exercise 4: RAM with Reset Initialization

Design a RAM that:

- Clears all memory locations on reset

Tasks:

- Use for loop for initialization
- Reset should be **synchronous or asynchronous**
- Verify all locations become 0

Exercise 5: Parameterized RAM

Design a **generic RAM module**.

Tasks:

- Use parameters:
    - DATA_WIDTH
    - ADDR_WIDTH
- Make memory scalable
- Test for at least 2 configurations (e.g., 8×8, 32×16)

ROM Design Exercises

Exercise 6: Basic ROM using Case Statement

Design a **4×8 ROM**.

Tasks:

- Use case statement
- Map fixed values to addresses
- Output should change based on input address

Exercise 7: ROM using Initial Block

Design an **8×8 ROM**.

Tasks:

- Use initial block to preload values
- Store any pattern (e.g., multiplication table)

Exercise 8: ROM for Code Conversion

Design ROM for **BCD to 7-segment display decoder**.

Tasks:

- Input: 4-bit BCD
- Output: 7-bit segment pattern
- Store mapping inside ROM

Exercise 9: ROM using File Initialization

Design ROM initialized from a file.

Tasks:

- Use $readmemb or $readmemh
- Load data from external file
- Verify correct loading in simulation

Exercise 10: Parameterized ROM

Design scalable ROM.

Tasks:

- Use parameters for size
- Make content configurable
- Verify with different sizes

Dual-Port RAM Exercises

Exercise 11: Basic Dual-Port RAM

Design a **dual-port RAM** with:

- Two independent ports (A and B)

Tasks:

- Each port has: addr, din, dout, we
- Allow simultaneous read/write

Exercise 12: Simultaneous Write Conflict

Design dual-port RAM and handle:

Tasks:

- Both ports write to **same address**
- Define behavior:
    - Priority (Port A or B)
    - OR undefined

Exercise 13: Dual-Port RAM with Separate Clocks

Design RAM with:

- Port A → clk1
- Port B → clk2

Tasks:

- Handle independent clock domains
- Analyze timing issues

Exercise 14: Dual-Port RAM (One Read, One Write)

Design RAM where:

- Port A → write-only

- Port B → read-only

Tasks:

- Ensure no conflict
- Verify simultaneous operation

Exercise 15: Parameterized Dual-Port RAM

Design scalable dual-port RAM.

Tasks:

- Use parameters for:
    - Data width
    - Address width